

00-environment-setup

Propietario	© Carlos Roberto
Etiquetas	

Script 01 - Creación de Base de Datos con Filegroups

```
-- =====  
--  
-- P02 - DATA GOVERNANCE ARCHITECTURE  
-- TRANSTRACK Logistics & Freight Transportation  
-- Week 00: Environment Preparation  
-- Script 01: Database Creation with Filegroups  
-- =====  
--  
--  
  
USE master;  
GO  
  
IF EXISTS (SELECT name FROM sys.databases WHERE name = N'TRANSTRACK')  
BEGIN  
    ALTER DATABASE TRANSTRACK SET SINGLE_USER WITH ROLLBACK IMMEDIATE;  
    DROP DATABASE TRANSTRACK;  
END  
GO  
  
-- Three filegroups aligned with workload tiers:  
-- PRIMARY - Transactional tables (OLTP)  
-- ANALYTICS - Dimensional models (read-heavy)  
-- ARCHIVE - Historical telemetry (bulk, retention-managed)
```

```

CREATE DATABASE TRANSTRACK
ON PRIMARY
(
    NAME = N'TRANSTRACK_Primary',
    FILENAME = N'C:\SQLData\TRANSTRACK_Primary.mdf',
    SIZE = 256MB,
    MAXSIZE = 2GB,
    FILEGROWTH = 128MB
),
FILEGROUP ANALYTICS
(
    NAME = N'TRANSTRACK_Analytics',
    FILENAME = N'C:\SQLData\TRANSTRACK_Analytics.ndf',
    SIZE = 128MB,
    MAXSIZE = 1GB,
    FILEGROWTH = 64MB
),
FILEGROUP ARCHIVE
(
    NAME = N'TRANSTRACK_Archive',
    FILENAME = N'C:\SQLData\TRANSTRACK_Archive.ndf',
    SIZE = 512MB,
    MAXSIZE = 10GB,
    FILEGROWTH = 256MB
)
LOG ON
(
    NAME = N'TRANSTRACK_Log',
    FILENAME = N'C:\SQLLog\TRANSTRACK_Log.ldf',
    SIZE = 256MB,
    MAXSIZE = 2GB,
    FILEGROWTH = 128MB
);
GO

-- FULL recovery: financial data + legal evidence require poi

```

```
nt-in-time restore
ALTER DATABASE TRANSTRACK SET RECOVERY FULL;
GO
```

The screenshot shows the 'Messages' window in SQL Server Enterprise Manager. It displays the execution details for the commands entered in the previous block. The messages include 'SQL Server parse and compile time:', 'SQL Server Execution Times:', and 'Table 'sysdbsreg'. Scan count 0, logical reads 2, physical reads 1, page server reads 0, read-ahead reads 0, page server read-ahead reads 0, lob logical reads 0, lob physical reads 0'. The status bar at the bottom indicates 'Query executed successfully.' and 'No issues found'.

Script 02 - Environment Verification

```
-- =====
-- Script 02 - Environment Verification
-- =====

USE TRANSTRACK;
GO

-- Verify database properties
SELECT
    name,
    recovery_model_desc,
    collation_name,
    compatibility_level
FROM sys.databases
WHERE name = N'TRANSTRACK';
GO
```

```

-- Verify filegroups and file sizes
SELECT
    fg.name                AS filegroup_name,
    df.name                AS logical_name,
    df.physical_name       AS physical_path,
    df.size * 8 / 1024     AS size_mb
FROM sys.filegroups fg
JOIN sys.database_files df ON fg.data_space_id = df.data_space_id
ORDER BY fg.name;
GO

-- Verify server config baseline for 16GB machine
SELECT
    name,
    value_in_use
FROM sys.configurations
WHERE name IN (
    'max degree of parallelism',
    'cost threshold for parallelism',
    'max server memory (MB)'
)
ORDER BY name;
GO

```

Results

Messages

	name	recovery_model_desc	collation_name	compatibility_level
1	TRANSTRACK	FULL	SQL_Latin1_General_CP1_CI_AS	160

	filegroup_name	logical_name	physical_path	size_mb
1	ANALYTICS	TRANSTRACK_Analytics	C:\SQLData\TRANSTRACK_Analytics.ndf	128
2	ARCHIVE	TRANSTRACK_Archive	C:\SQLData\TRANSTRACK_Archive.ndf	512
3	PRIMARY	TRANSTRACK_Primary	C:\SQLData\TRANSTRACK_Primary.mdf	256

	name	value_in_use
1	cost threshold for parallelism	5
2	max degree of parallelism	8
3	max server memory (MB)	2147483647

Query executed successfully.

c13\lab_db (16.0 RTM)carlos_roberto (53)TRANSTRACK00:00:00Row: 1, Col: 17 rows

Verificación MDS

```
-- =====
--
-- P02 - Week 00: Script 03 - Master Data Services Verificati
on
-- =====
--
-- Check if MDS database exists
SELECT name, state_desc
FROM sys.databases
WHERE name LIKE '%MDS%';
GO

-- =====
--
-- MANUAL STEPS:
-- 1. SQL Server Configuration Manager -> verify SQL Server A
gent is running
-- 2. Open Master Data Services Configuration Manager
--    - If not installed: SQL Server installer -> Add Feature
s -> MDS
```

```
--      - If installed: Create database MDS_TRANSTRACK, create
web app, associate
-- 3. Verify: http://localhost/MDS opens Master Data Manager
-- =====
=====

PRINT 'MDS check complete. Follow manual steps if not configu
red.';
GO
```

Note: There is currently no MDS configuration in the selected environment, so it will be part of the implementation if required.

Python connectivity test

```
import pyodbc
from config import SERVER, DATABASE, DRIVER

def get_connection():
    conn_str = (
        f"DRIVER={{DRIVER}};"
        f"SERVER={SERVER};"
        f"DATABASE={DATABASE};"
        "Trusted_Connection=yes;"
        "Connection Timeout=10;"
    )
    return pyodbc.connect(conn_str)
```

```
# =====
=====
# P02 - Week 00: verify_environment.py
# Requires: config.py, connection.py
# =====
```

=====

```
import sys
import pyodbc
from connection import get_connection

print(f"Python : {sys.version}")
print(f"pyodbc : {pyodbc.version}")

try:
    conn = get_connection()
    cur = conn.cursor()

    cur.execute("SELECT @@VERSION")
    print(f"SQL      : {cur.fetchone()[0].split(chr(10))[0]}")

    cur.execute("""
        SELECT name, recovery_model_desc, collation_name
        FROM sys.databases WHERE name = DB_NAME()
    """)
    row = cur.fetchone()
    print(f"DB      : {row.name}")
    print(f"Recovery: {row.recovery_model_desc}")
    print(f"Collation: {row.collation_name}")

    cur.execute("""
        SELECT fg.name, df.size * 8 / 1024 AS size_mb
        FROM sys.filegroups fg
        JOIN sys.database_files df ON fg.data_space_id = df.d
ata_space_id
        ORDER BY fg.name
    """)
    print("Filegroups:")
    for r in cur.fetchall():
        print(f"  {r.name:20s} {r.size_mb} MB")
```

```
cur.close()
conn.close()
print("\nConnection: OK")

except pyodbc.Error as e:
    print(f"\nConnection FAILED: {e}")
    sys.exit(1)
```



The screenshot shows a PowerShell terminal window with the following content:

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS
(venv) PS C:\data-governance-architecture\00-environment-setup> python connectivity_test.py
Python : 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]
pyodbc : 5.3.0
SQL : Microsoft SQL Server 2022 (RTM) - 16.0.1000.6 (X64)
DB : TRANSTRACK
Recovery: FULL
Collation: SQL_Latin1_General_CP1_CI_AS
Filegroups:
ANALYTICS 128 MB
ARCHIVE 512 MB
PRIMARY 256 MB

Connection: OK
(venv) PS C:\data-governance-architecture\00-environment-setup>
```

The terminal window has a title bar with standard Windows icons and a status bar at the bottom showing "Ln 7, Col 28", "Spaces: 4", "UTF-8", "CRLF", "Python", and "venv (3.14.2)".